

Annotated Bibliography Updated for Algorithm 2

Comparing Energy Costs of Parallel DNN Training Techniques

Note: Papers [1]-[30] remain from the March 18 submission. This update adds papers [31]-[35] covering quantization precision levels (BF16, INT8) and Green AI methodology, per Professor Prasad's feedback to include a range of quantization techniques for Algorithm 2.

8. Algorithm 2 Extended Quantization & Precision Techniques

[31] Kalamkar, D., et al. "A Study of BFLOAT16 for Deep Learning Training." arXiv, 2019.
<https://arxiv.org/abs/1905.12322>

This is the foundational study of BF16 as a training format, showing that it achieves the same accuracy as FP32 across image classification, speech, language modeling, and recommendation tasks without any hyperparameter changes. The key advantage over FP16 is that BF16 has the same exponent range as FP32, which means it never needs loss scaling. For Algorithm 2 this is directly relevant BF16 is one of the quantization levels being compared, and this paper explains exactly why it behaves differently from FP16 despite being the same memory size. Originally developed at Intel and now the default training format on A100/H100 GPUs.

[32] "Gradient Distribution-Aware INT8 Training for Neural Networks." ScienceDirect, 2023.
<https://www.sciencedirect.com/science/article/abs/pii/S0925231223003922>

This paper addresses the core challenge of INT8 training gradient instability by proposing a dynamic segmentation quantization scheme that adjusts quantization parameters based on the distribution of gradients during training. The authors show that naive INT8 quantization causes training to diverge, but their approach stabilizes it. Directly relevant to Algorithm 2 because INT8 is one of the precision levels being tested, and understanding why INT8 is harder than FP16 or BF16 helps explain the accuracy tradeoffs in the results. The paper also explicitly connects INT8 training to energy reduction and carbon footprint, framing it within the same Green AI motivation as this project.

[33] "Green AI Techniques for Reducing Energy Consumption in AI Systems." ScienceDirect, 2025. <https://www.sciencedirect.com/science/article/pii/S2590005625002796>

This 2025 review covers the full landscape of energy reduction techniques for AI including quantization, pruning, distillation, and hardware co-design, with a focus on reported energy savings across studies. Notably, it reports that INT8 post-training quantization achieves 21-54% energy savings at inference compared to FP32 baselines. For this project, this paper serves as the measurement benchmark if INT8 delivers that range of savings at inference, how much does it save during training? The paper also discusses standardized power benchmarks like MLPerf Power which could be relevant for validating energy measurements from nvidia-smi.

[34] "Evaluating Quantized LLMs for Energy Efficiency, Output Quality." ACM Transactions on Internet of Things, 2025. <https://dl.acm.org/doi/pdf/10.1145/3767742>

This paper systematically compares FP16, INT8, and INT4 quantization on LLMs deployed at the edge, measuring energy efficiency, inference speed, and accuracy tradeoffs across multiple model families. The authors find that INT8 offers the best balance between efficiency and accuracy, while INT4 reduces power further but at significant accuracy cost. For Algorithm 2, this provides

direct benchmarks for what accuracy-energy tradeoffs to expect across precision levels, and the methodology for integrating hardware-level energy profiling with model benchmarking is a useful reference for the measurement design in this project.

[35] "Energy-Aware Deep Learning for Real-Time Video Analysis through Pruning, Quantization, and Hardware Optimization." Journal of Real-Time Image Processing, 2025.

<https://link.springer.com/article/10.1007/s11554-025-01703-0>

This 2025 paper evaluates the combined effect of pruning and quantization on energy consumption using ResNet and YOLO architectures on both A100 GPUs and Jetson Nano edge devices the same ResNet architecture used in this project. The authors find that pruning alone reduces energy by 35% and quantization adds another 18% on top, with minimal accuracy loss. Directly useful for Algorithm 2 because it provides energy savings numbers specifically for ResNet quantization that can serve as a comparison baseline for this project's measured results. The use of NVIDIA Nsight Systems and nvprof for profiling alongside nvidia-smi is also methodologically relevant.

Classification Scheme

The 35 papers in this bibliography are organized into a classification scheme that maps directly onto the three algorithms of this project:

Class 1 Data Parallelism Frameworks (Algorithm 1)

Primary: [1] PyTorch DDP, [3] AxoNN, [2] Horovod, [5] IEEE MICRO 2025

Secondary: [4] Parameter Server, [6] Fine-grained Parallelism, [18] GPipe, [19] PipeDream, [20] Megatron-LM

Class 2 Precision & Quantization (Algorithm 2)

Primary: [7] Mixed Precision FP16, [31] BF16, [32] INT8, [8] Energy Impact of Mixed Precision

Secondary: [9] Quantization Hardware Survey, [33] Green AI Techniques, [34] Quantized LLMs, [35] Pruning + Quantization

Class 3 Memory Optimization & Distillation (Algorithm 3)

Primary: [10] ZeRO, [11] ZeRO-Infinity, [14] Knowledge Distillation

Secondary: [12] Distillation + Quantization, [16] Memory Control for LLMs, [17] ZeRO-Offload

Class 4 Environmental Impact & Motivation

Primary: [21] Carbon/Water Footprints, [23] Strubell Green AI, [22] Cornell Data Center Roadmap

Secondary: [15] Energy-Aware DL, [33] Green AI Techniques 2025

Class 5 Prasad Lab & Foundational Papers

Primary: [24] Mahmud Dissertation, [25] AxoNN Sparsity, [26] Democratizing AI

Secondary: [27] LeCun 1989, [28] ResNet, [29] AlexNet, [30] HOGWILD!